



IHMF-PD: An interpretable holistic multimodal framework for classifying Parkinson's disease severity[☆]

Nutan Singh^a,^{*}, Priyanka Tripathi^a, Abhijeet Kumar Kohat^b

^a Department of Computer Applications, National Institute of Technology Raipur, G.E. Road, Raipur 492010, Chhattisgarh, India

^b Department of Neurology, DKSPGI (Dau Kalyan Singh Postgraduate Institute), Raipur 492010, Chhattisgarh, India

ARTICLE INFO

Keywords:

Parkinson's disease
Wearable sensors
Feature-driven
Severity
Machine learning
Interpretability

ABSTRACT

Classifying the severity of Parkinson's Disease (PD) remains one of the most challenging tasks in neuroscience. Accurate detection of PD progression is essential for timely and effective treatment. To address this challenge, we propose an Interpretable Holistic Multimodal Framework (IHMF) for two-stage severity classification of PD. In this work, we collected real-time hand tremor data from Parkinson's patients based on a 9-axis IMU wrist sensor (Inertial Measurement Unit), along with clinical data such as the society-sponsored review of the Unified PD Rating Scale (MDS-UPDRS-I, II, III), Hoehn and Yahr (H&Y), and PD Questionnaire (PDQ-39) scores by neurologists. For severity ground truth labeling, we used the MDS-UPDRS and H&Y score. Sensor data captured tremor information from both the postural and rest states of the hands, creating a holistic, multimodal framework to quantify the severity of PD with high precision. We used optimized machine learning (ML) models for severity classification. The voting classifier demonstrated superior performance, achieving an accuracy of 90.78% for the severity of tremors and 99.48% for the severity of PD, outperforming other classifiers. In addition, we used model interpretability techniques to ensure the transparency of clinical evaluation. These interpretability tools provide insights into the model's decision-making process, enabling neurologists to validate and trust the predictions. This study highlights the potential of integrating multimodal sensor data and an optimized model for accurate and interpretable prediction.

1. Introduction

Parkinson's Disease (PD) is a progressive neurodegenerative disorder that predominantly affects individuals over the age of 60, leading to significant motor and non-motor impairments. Among motor symptoms, tremor is one of the most prevalent and earliest manifestations of PD, often serving as a critical diagnostic marker [1]. During the past 25 years, the global prevalence of PD has doubled. With the projected increase in the ageing population by 34% by 2050, there is an urgent need for advanced tools to diagnose and monitor the disease effectively [2].

Tremors in PD are categorized into two types: resting and postural hand tremors. A resting tremor occurs when a body segment is at rest and typically exhibits a frequency range of 3.5 to 7.5 Hz. In contrast, postural tremor occurs when a body segment maintains a posture against gravity, such as holding a cup, and usually ranges from 4 to 12 Hz [3]. Although resting tremor is a hallmark symptom of PD, it

can decrease during posture or movement in the early stages but persist during these states in the advanced stages of the disease.

Recent advances in Artificial Intelligence (AI) and wearable sensor technologies have shown promise in allowing accurate and objective assessment of symptoms of PD, including the severity of tremors and the progression of the disease. Several studies have proposed computational methods for tremor detection and quantification, including time-domain, spectral, time-frequency, and non-linear. Wearable devices such as Kinesia™, Perform System, and Kinetigraph have also been developed to measure motor symptoms in PD patients [4]. However, accurately diagnosing the severity of PD tremors and the overall disease stage remains a significant challenge in clinical practice. PD diagnosis has seen significant advances through computational analysis of various biomarkers, including medical imaging, vocal patterns, gait dynamics, and hand movement analysis. However, a significant gap remains in terms of the reliability and generalizability of these models [5]. Most

[☆] This work is a thoroughly revised and extended version of the conference paper titled – "WTM-PD: Wearable IMU-Based Model for Real-Time Parkinson's Tremor Analysis in Rest and Postural State", 5th International Conference on Computer Vision and Robotics (CVR 2025), India. Sections 3.4, 3.5, 4.2 and 4.3, involving results related to the holistic approach to severity quantification, are new additions to the conference version.

^{*} Corresponding author.

E-mail addresses: nsingh.phd2021.mca@nitrr.ac.in (N. Singh), ptripathi.mca@nitrr.ac.in (P. Tripathi), drabhijeetkk@gmail.com (A.K. Kohat).

approaches rely on a single modality, which limits their ability to accurately classify PD, especially in early stages where motor symptoms may not be prominent. This can lead to biased or incorrect severity predictions, as some patients with PD may not exhibit visible motor symptoms in the initial stages.

To address these limitations, we propose a novel two-stage Interpretable Holistic Multimodal Framework for Parkinson's Disease (IHMF-PD) that synergistically combines quantitative sensor data with comprehensive clinical assessments. In the first stage, Inertial Measurement Unit (IMU) objectively quantify tremor characteristics during both resting and postural states. The second stage integrates these motor metrics with clinician-rated Movement Disorder Society – Unified Parkinson's Disease Rating Scale (MDS-UPDRS)-I, II, III and Hoehn and Yahr (H&Y) staging scale (scores 1–4) (comprehensively assessing non-motor and motor symptoms) and patient-reported Parkinson's Disease Questionnaire – 39 (PDQ-39) quality-of-life measures. This multimodal approach not only improves diagnostic accuracy across disease stages but also provides clinically interpretable insights by correlating objective sensor measurements with established clinical benchmarks. By bridging the gap between quantitative signal analysis and holistic patient assessment, our framework offers a more robust solution for PD severity classification that addresses the limitations of current unimodal systems.

This work represents an extended version of our previous conference paper, with significant enhancements to the experiment. The primary objective of this work is to accurately predict tremor and PD severity, ensuring that misclassification of other PD symptoms as tremor is minimized to avoid inaccurate patient assessments. The main contributions of this work are as follows:

1. We collected real-time hand tremor data (resting and postural states) from PD patients using wearable IMU sensors, combined with clinical evaluations (MDS-UPDRS and PDQ-39 scores) validated by neurology experts.
2. We developed a novel two-stage framework to classify tremor severity (using IMU data) and PD severity (by integrating tremor results with clinical data), enabling precise and holistic PD assessment.
3. We employed optimized model and feature selection techniques to identify discriminative features, achieving accurate classification of tremor and PD severity.
4. Interpret the model decisions.

There are four sections in the remainder of the paper. The most recent research on PD diagnosis is provided in Section 2. The methods utilized to construct the suggested structure are explained in Section 3. Section 4 discusses the proposed framework's experimental results, limitations, and comprehensive analysis. We have concluded our work in Section 5.

2. Related work

Recent studies have explored wearable sensors and machine learning for PD tremor detection and severity classification. Using a customized quantitative assessment approach based on MDS-UPDRS criteria, [3] could quantify postural tremor and rest with reasonable accuracy. To estimate PD hand tremors, [6] created a reliable categorization approach with a 100% association with UPDRS scores. A continuous, objective assessment system for PD symptoms during daily living was created by [7], who demonstrated 83% accuracy for clinically significant parameters. Using wrist-worn accelerometers, [4] evaluated novel preprocessing techniques and achieved an error lower than 5% in tremor percentage estimation. A classification strategy for identifying PD and healthy control participants was presented by [8], showing potential clinical value with 89% accuracy. In 2020, [9] concentrated on utilizing LSTM to differentiate between ET and PSD tremors, attaining 90% accuracy and considering adaptability for tracking daily

tremors. According to [10], a pilot study used video data analysis to separate people with PD from subjects with essential tremors. [11] used a real-time detecting robot with a 93% classification accuracy to establish a dependable diagnosis method for early-stage PD. With three classes, [12] measured the severity of PD and used CNN and AdaBoost to get 89% sensitivities. Improved the classification of PD tremor severity, [13] utilizing ANN-MLP to achieve 93.81% accuracy. They recommended further research with various resampling methods.

A wearable device used [14] reported an accuracy of $85\% \pm 15\%$ in their use of wrist-worn accelerometers to diagnose PD early; they recommended a long-term investigation. Using wearable sensors, [15] developed a CNN-based classification system that achieved 92.4% accuracy in classifying PSD and HC individuals as well as differentiating between ET and PSD tremors. In exploring mobile-based apps for real-life monitoring and help, [16] found that Bi-LSTM with an attention mechanism could achieve high accuracy. They have developed mobile applications integrating accelerometers for real-time monitoring of PD patients. They employed a Bidirectional Long Short-Term Memory (Bi-LSTM) model with an attention mechanism to analyze movement data, achieving impressive accuracies of 98% in distinguishing PD patients from healthy controls and 99% in detecting PD-specific movements.

To improve vision-based estimation methods for measuring the intensity of tremors in PD, [17] developed a new model using video recordings' MDS-UPDRS score. The Global Temporal-difference Shift Network (GTSN) is a model that uses a Global Shift Module (GSM) to intensify and focus on small tremors. It has residual networks (ResNet) and Eulerian video magnification (EVM). Tests showed that they could accurately predict the degree of tremors with 90.6% accuracy for hand rest tremors, 85.9% for leg tremors, 89.0% for jaw tremors, and 84.9% for postural tremors. [18] introduced vision-based alternatives with interpretable AI, achieving 85% accuracy for multi-class classification. [19] achieved strong binary ($F1 = 0.938$) and multiclass ($F1 = 0.867$) PD classification using wrist-worn IMUs and sensor fusion, though their left-hand dominance finding warrants further investigation. [20] demonstrated the challenges of real-world deployment, with lab-to-home accuracy drops (e.g., tremor: $83\% \rightarrow 63\%$) highlighting environmental variability's impact. While ensemble models excelled in controlled exercises, this work revealed ankle sensors' limited utility (only aiding dyskinesia detection). Both studies emphasize the need for larger cohorts, multimodal data integration, and longitudinal validation to bridge the gap between algorithmic performance and clinical applicability. With this procedure, PD tremors can be clinically assessed dependably and effectively.

Numerous studies focus on developing and evaluating systems for the remote measurement and analysis of motor symptoms utilizing a range of inertial sensors depicted in Table 1. This table compares our collected dataset with another private or public dataset. In the existing research work, we found multiple studies that suggest the multimodal concept of clinical data for diagnosing PD is more accurate. Despite these advancements, several gaps remain: (1) limited dataset sizes, (2) lack of multimodal data integration, (3) insufficient focus on explainability, and (4) the need for real-time, continuous monitoring.

To address these challenges, we propose a IHMF-PD that integrates IMU sensor data with clinical assessments, leveraging ensemble learning and explainable AI techniques. The two most often used technologies were accelerometers and gyroscopes. Although many classification methods were used in this study, there was discussion over the most effective approach for assessing performance development. Usually, only small datasets were examined; hence, additional evidence of the applicability of ML is required [25]. Previous studies have explored various techniques for gathering data from PD patients [26,27]. However, further study must identify the most accurate methods for classifying PD severity [22]. By addressing the limitations of existing PD monitoring systems, this study aims to contribute to developing more effective and efficient tools for PD diagnosis and management. Integrating wearable sensor data, advanced signal processing, and machine learning algorithms holds significant potential for improving clinical decision-making and patient outcomes.

Table 1

Comparative analysis of our dataset with other datasets for wearable devices in PD diagnosis.

Authors and Year	Wearable Device	Measure	Diagnose Year	Sample Size	Pose				Age (y)	M	F	MD-UPDRS				PDQ-39
					R_L	R_R	P_L	P_R				I	II	III	H&Y scale	
San-Segundo R et al. 2020 [4]	Three-axis (X, Y and Z)	Acceleration	2–5 years	12	✓	✓	✓	✓	62–85	–	–	–	–	✓	–	–
Jeon H et al. 2017 [21]	Triaxis accelerometer sensor	The acceleration and angular velocity of hand tremors	–	85	✓	✓	–	–	65.96 ± 9.19	41	44	–	–	✓	–	–
López-Blanco R et al. 2019 [22]	Smartwatch3 (SW3) Sonyc worn at the wrist	Acceleration and velocity signals	1 year	22	✓	✓	–	–	72 ± 7.6	13	9	–	–	✓	–	–
Kostikis N et al. 2015 [23]	iPhone 4S	Acceleration and velocity signals	–	23 PD, 20 HC	✓	✓	✓	✓	70.91 ± 11.78	20	23	–	–	✓	–	–
Patel S et al. 2009 [24]	SHIMMER	Uniaxial accelerometer sensors	–	12	✓	✓	✓	✓	46–75	–	–	–	–	✓	–	–
Sigcha L et al. 2021 [12]	mHealth mobile and wearable application	Accelerometer	–	18	✓	✓	–	–	64.9 ± 7.6	8	10	–	–	✓	✓	–
Colum Crowe et al. 2024 [12]	Wrist-worn IMU sensor, Ankle-worn IMU sensor	Accelerometer and gyroscope	–	24	✓	✓	–	–	70.71 ± 8.50	16	8	–	–	✓	✓	–
Mohammed Hammoud et al. 2025 [19]	Wrist-worn IMU sensors (left and right hands)	Accelerometer	–	PD = 55, HC = 30	✓	✓	–	–	–	–	–	–	–	✓	✓	–
Our dataset	Nine axial accelerometers	Acetometer, gyroscope, magnetometer	1-year	PD = 58, HC = 20	✓	✓	✓	✓	57.48 ± 12.11	14	44	✓	✓	✓	✓	✓

Note: We indicate whether the investigated research includes these algorithm or technique categories using “✓” and “–”. A “✓” signifies inclusion, while a “–” signifies exclusion. (R_L: Rest Left, R_R: Rest Right, P_L: Postural Left, P_R: Postural Right).

3. Methodology

This section describes the complete workflow of the proposed IHMF-PD framework, starting from data acquisition, followed by preprocessing, feature extraction, model training, and interpretability analysis. Each step is explained in the subsequent subsections. Fig. 1 depicts the proposed framework outlining data acquisition, preprocessing, feature extraction, and two-stage classification for tremor and overall PD severity assessment. The proposed framework operates in two stages. First, it classifies tremor severity, a key early symptom of Parkinson's disease, to guide medication and improve patient outcomes. Second, it assesses overall PD severity using a holistic dataset combining sensor data and clinical assessments, including MDS-UPDRS Parts I-III scores, H&Y staging, and PDQ-39 scores. This integrated approach ensures precise and comprehensive PD severity predictions, as illustrated in Fig. 2. The framework consists of six phases: (1) Clinical Tasks for Data Collection, (2) Data Preprocessing, (3) Feature Extraction, (4) Feature Selection, (5) Severity Assessment, and (6) Significant Features Selection. Experimental validation is discussed in the following section.

3.1. Data acquisition

3.1.1. Ethical considerations

This study was approved by the Institutional Ethics Committee of the National Institute of Technology Raipur, Chhattisgarh (Approval

No. NITRR/IEC/2023/06). All procedures were conducted in accordance with the Declaration of Helsinki, and informed consent was obtained from all participants. Data collection was carried out at the Department of Neurology, Dau Kalyan Multispecialty Hospital, Raipur, Chhattisgarh, India, with ethical approval obtained from both the hospital's medical ethics committee and the ethics committee of the National Institute of Technology Raipur.

3.1.2. Device configuration

This study employed a high-performance IMU sensor [28] to capture accelerations, gyroscopic data, angles, magnetic field, and quaternion measurements. The device connects to a laptop via a USB port and provides a user interface for configuration. All data were recorded in .txt files. The sensor's output rate and measurement range are adjustable from 0.2 to 200 Hz. We set the output rate to 50 Hz and the baud rate to 9600 for our experiments utilizing the 9-axial system algorithm. The device incorporates a Kalman filter algorithm, suitable for capturing tremor frequencies typically ranging from 3–12 Hz for rest tremors and 4–8 Hz for postural tremors. A higher sampling rate was deemed unnecessary, as it would generate excessive data without adding meaningful information [3].

This study included 78 participants: 58 PD patients and 20 HC. Participants were selected based on the presence of idiopathic PD with tremor symptoms. The mean age of the participants was 57.48 ± 12.11 years, as detailed in Table 3. A balanced sample of PD and HC is not required, as the study does not classify PD or healthy control

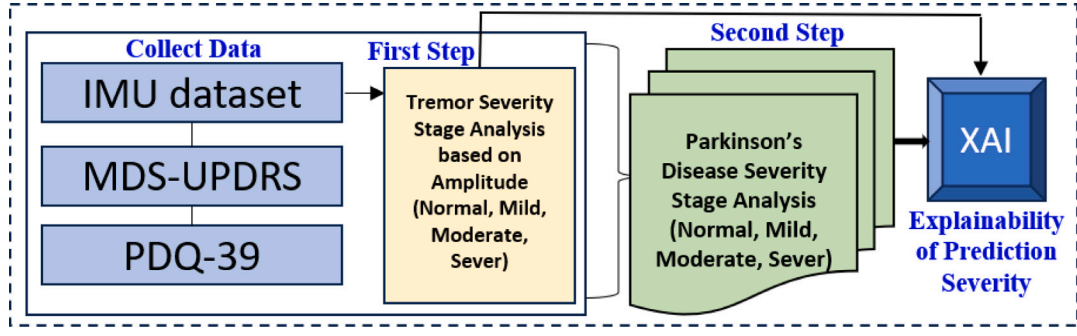


Fig. 1. Proposed framework steps.

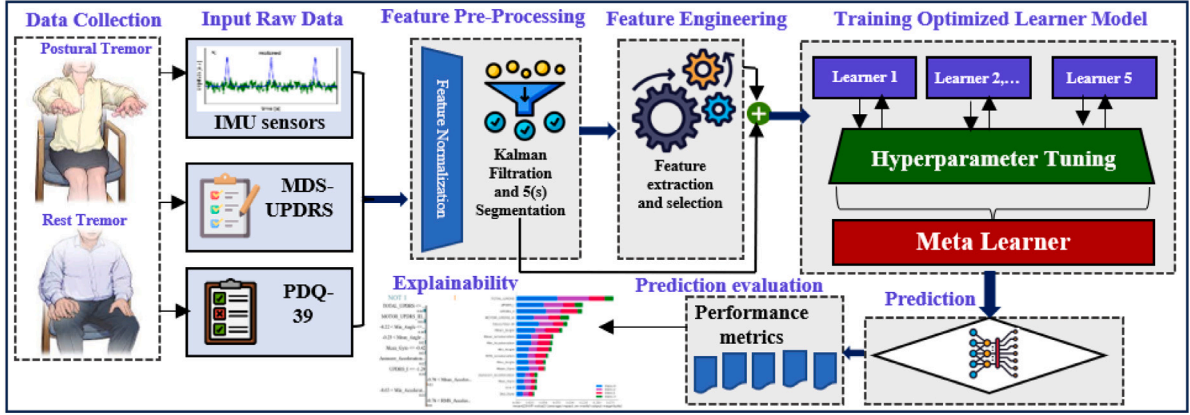


Fig. 2. Proposed framework structure for severity assessment.

Table 2
Technical configuration of sensor.

Specification	Details
Model/Brand	WT901C/WitMotion
Input voltage	TTL: 3.3–5 V; RS232/RS485: 5–36 V
Current	<25 mA
Interface	TTL/RS232/RS485 (Modbus)
Output data	3-axis: Acceleration, Gyroscope, Angular Velocity, Magnetic Field, Quaternion
Measuring range	Acceleration: $\pm 16g$; Gyroscope: $\pm 2000^\circ/s$; Angle: $X/Z \pm 180^\circ$, $Y \pm 90^\circ$
Resolution	Acceleration: 0.0005g; Gyroscope: $0.61^\circ/s$
Stability	Acceleration: 0.01g
Angle accuracy	XY Axis: Static 0.05° , Dynamic 0.1°
Output frequency	0.2–200 Hz (Default: 10 Hz)
Baud rate	4800–230 400 (Default: 9600)
Size	$52 \times 36 \times 15$ mm ($2.05'' \times 1.41'' \times 0.59''$)
Installation	Vertical or horizontal (optional)
Net Weight	20 g
Axial Definition	Cartesian Coordinates (Right-hand Rule)

patients but instead categorizes them into normal (no tremor), mild, moderate, and severe classes. The methodology presented in this study focuses on tremor severity in PD patients using IMU sensor data and PD severity using a holistic, multimodal approach (IMU + MDS-UPDRS + PDQ-39) [29].

3.1.3. Clinical tasks for data collection

In the MDS-UPDRS, the assessment of rest and postural tremors in the hands falls under Part III: Motor Examination, specifically Postural Tremor of the Hands (Item 3.15) and Rest Tremor Amplitude (Item

3.17). We collect records for both hands separately in both rest and postural states. Fig. 3(a) and Table 2 illustrate the data samples and configuration.

Task 1: Postural Tremor (Item 3.15): Participants, while seated, extended their arms forward with hands pronated at shoulder level and held this position for at least 10 s to assess the presence and amplitude of postural tremor.

Task 2: Rest Tremor (Item 3.17): Participants rested both hands on their lap while seated, closed their eyes, and remained in this position for at least 10 s to allow observation of rest tremor.

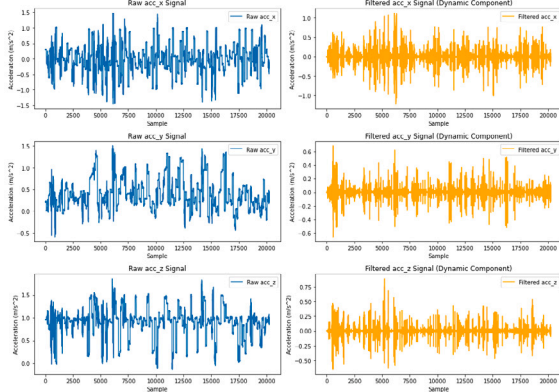
Patients with PD were assessed in the OFF-levodopa state, defined as a period of at least 12 h after their last medication dose. During each assessment task, 10 s of motion data were recorded from the body side exhibiting more severe tremor symptoms [30]. Additionally, we collected PDQ-39 scores from each patient in the ON-medication state, focusing on mobility-related functionality. This combination of quantitative tremor data and patient-reported outcomes strengthens the key contribution of our work: an integrated framework for assessing PD severity. Our dataset analysis revealed that tremor onset occurred more frequently in the left hand among most patients.

3.2. Data preprocessing

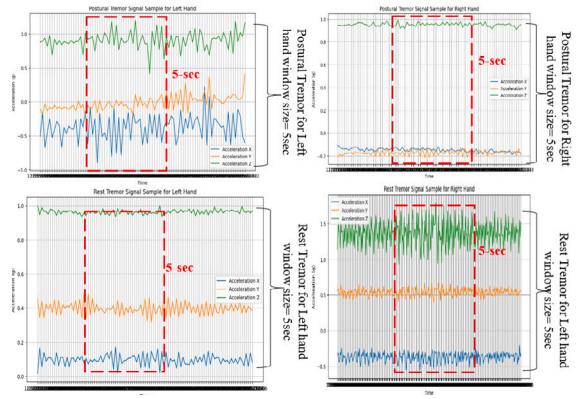
The raw IMU data consists of time-series measurements from 9-axis sensors, which include accelerometer readings $\mathbf{a} = [a_x, a_y, a_z]$, gyroscope readings $\mathbf{w} = [w_x, w_y, w_z]$, and angular positions $\boldsymbol{\theta} = [\theta_x, \theta_y, \theta_z]$. The raw dataset X_{raw} is represented as:

$$X_{\text{raw}} = \{\text{Time}, \mathbf{a}, \mathbf{w}, \boldsymbol{\theta}\}$$

where Time represents the timestamp for each measurement. The pre-processing begins with data cleaning, where non-numeric columns,



(a) Kalman filtering on tremor signal.



(b) Segmentation into fixed 5-second windows.

Fig. 3. (a) Kalman filtering of tremor signal. (b) Segmentation of filtered signal into fixed windows.**Table 3**

Clinical and Demographic Information About the Patients.

Attributes	Mean $\pm$ SD/Values
Age	57.48 $\pm$ 12.11
Gender (Total)	PD = 58, HC = 20
Female	52
Male	26
Height (m)	5.37 $\pm$ 0.026
Weight (kg)	61.72 $\pm$ 11.42
Duration (years)	3.51 $\pm$ 2.76
Clinical scores at recruitment	
MDS-UPDRS I	11.48 $\pm$ 5.44
MDS-UPDRS II	9.75 $\pm$ 6.0
MDS-UPDRS III	39.05 $\pm$ 15.86
Hoehn & Yahr	2.01 $\pm$ 0.47
Other clinical scores	
PDQ-39	42.51 $\pm$ 19.95

Arm Postural/Rest tremor for both hands.

such as timestamps and sensor metadata, are discarded. Missing or invalid data points are removed, resulting in a cleaned dataset X_{clean} :

$$X_{\text{clean}} = \{\mathbf{a}, \mathbf{w}, \theta\}$$

3.2.1. Filtration

Following this, the Kalman filter $\mathcal{K}$ is applied to the cleaned data to reduce measurement noise. The Kalman filter provides an estimate $\hat{X}_{\text{filtered}}$ for each sensor reading $\mathbf{a}_k, \mathbf{w}_k, \theta_k$ as:

$$\hat{X}_{\text{filtered}} = \mathcal{K}(X_{\text{clean}})$$

This step smooths the raw data and mitigates noise from accelerometers and gyroscopes. Finally, the filtered data is segmented into fixed-length windows of 5 s, corresponding to $L = 250$ samples. The segmented data is represented as:

$$X_{\text{segment}} = \{\text{segment}_1, \text{segment}_2, \dots, \text{segment}_N\}$$

3.3. Feature extraction

This study performs feature extraction on tremor-related signals obtained from a 9-axis IMU sensor to derive meaningful statistical, frequency-domain, and non-linear patterns that characterize PD tremors' underlying patterns. Captures high-resolution, multi-dimensional motion data essential for characterizing PD tremors. We can comprehensively quantify tremor dynamics by extracting time-domain, frequency-domain, and nonlinear non-linear, enabling accurate classification and

severity prediction. The IMU sensor provides triaxial measurements for acceleration (a_x, a_y, a_z), angular velocity (w_x, w_y, w_z), and orientation angles ($\theta_x, \theta_y, \theta_z$). To reduce dimensionality and capture the overall magnitude of these signals, we compute the Euclidean norm for each triaxial measurement as follows:

$$\text{MaxAmplitude} = \sqrt{a_x^2 + a_y^2 + a_z^2}, \quad (1)$$

$$\text{GyroMagnitude} = \sqrt{w_x^2 + w_y^2 + w_z^2}, \quad (2)$$

$$\text{AngleMagnitude} = \sqrt{\theta_x^2 + \theta_y^2 + \theta_z^2}. \quad (3)$$

These magnitude-based features provide a consolidated representation of the triaxial signals, simplifying subsequent analysis while preserving essential information about tremor dynamics. The extracted features are categorized into time-domain, frequency-domain, and nonlinear.

3.3.1. Time-domain features

Time-domain features provide insights into the statistical properties of the tremor signals. For a given signal $x = \{x_1, x_2, \dots, x_N\}$, where N is the number of samples, the following features are computed:

- **Mean (μ):** Measures the central tendency of the signal:

$$\mu = \frac{1}{N} \sum_{i=1}^N x_i. \quad (4)$$

- **Standard Deviation (σ):** Quantifies the variability of the signal:

$$\sigma = \sqrt{\frac{1}{N} \sum_{i=1}^N (x_i - \mu)^2}. \quad (5)$$

- **Skewness (γ):** Describes the asymmetry of the signal distribution:

$$\gamma = \frac{\frac{1}{N} \sum_{i=1}^N (x_i - \mu)^3}{\sigma^3}. \quad (6)$$

- **Kurtosis (κ):** Measures the peakedness of the signal distribution:

$$\kappa = \frac{\frac{1}{N} \sum_{i=1}^N (x_i - \mu)^4}{\sigma^4}. \quad (7)$$

- **Root Mean Square (RMS):** Represents the overall energy of the signal:

$$\text{RMS} = \sqrt{\frac{1}{N} \sum_{i=1}^N x_i^2}. \quad (8)$$

- **Peak Count:** The number of local maxima in the signal, identified using a peak detection algorithm.
- **Autocorrelation** ($R(\tau)$): Assesses the periodicity and self-similarity of the signal:

$$R(\tau) = \frac{\sum_{i=1}^{N-\tau} (x_i - \mu)(x_{i+\tau} - \mu)}{\sum_{i=1}^N (x_i - \mu)^2}, \quad (9)$$

where τ is the time lag.

3.3.2. Frequency-domain features

Frequency-domain features are derived from the signal's power spectral density (PSD), computed using Welch's method. These features capture the spectral characteristics of PD tremors, which typically fall within the range of 3–12 Hz for rest tremors and 4–8 Hz for postural tremors. Key frequency-domain features include:

- **Dominant Frequency** (f_{dom}): The frequency with the highest PSD within the tremor range:

$$f_{\text{dom}} = \arg \max_{f \in [3, 12]} P(f), \quad (10)$$

where $P(f)$ is the PSD at frequency f .

- **Peak Power** (P_{peak}): The maximum PSD value within the tremor frequency range:

$$P_{\text{peak}} = \max_{f \in [3, 12]} P(f). \quad (11)$$

3.3.3. Nonlinear features

Non-linear is employed to capture the complexity and irregularity of the tremor signals. These features are particularly effective in characterizing the unpredictable nature of PD-related tremors. One such feature is **Fuzzy Entropy (FuzzyEn)**, which measures the signal complexity as follows:

$$\text{FuzzyEn}(m, r, n) = \ln \phi(m, r) - \ln \phi(m + 1, r), \quad (12)$$

where m is the embedding dimension, r is the tolerance, n is the fuzzy power, and $\phi(m, r)$ is the average similarity of signal segments of length m .

The extracted features are aggregated for each segment, resulting in a comprehensive feature set for PD tremor assessment.

3.4. Feature selection

After feature extraction, we perform feature selection to identify the most discriminative features for PD severity classification. Let $X \in \mathbb{R}^{n \times p}$ represent the feature matrix, where n is the number of samples and p is the number of extracted features. The target variable is denoted as $y \in \mathbb{R}^n$. We employ three feature selection methods to select the top 10 features:

1. **Filter Method:** Using the ANOVA F-test, we evaluate the relationship between each feature X_i and the target variable y . The F-score for feature X_i is computed as:

$$F(X_i, y) = \frac{\text{Variance between groups}}{\text{Variance within groups}}.$$

The top 10 features with the highest F-scores are selected.

2. **Wrapper Method:** Recursive Feature Elimination (RFE) with a logistic regression model is used to eliminate the least essential features iteratively. The process continues until only the top 10 features remain.

3. **Embedded Method:** Random Forest is used to compute feature importance based on Gini impurity reduction. The importance score for a feature X_i is:

$$\text{Importance}(X_i) = \frac{1}{N} \sum_{t=1}^N \Delta \text{Gini}(X_i, t),$$

where N is the number of trees and $\Delta \text{Gini}(X_i, t)$ the reduction in Gini impurity due to X_i in tree t . The top 10 features with the highest importance scores are selected.

To ensure robustness, we aggregate the results from the three methods. A feature is included in the final subset if selected by at least two of the three methods. The final feature subset $X_{\text{final}} \in \mathbb{R}^{n \times 10}$ is used for subsequent model training and evaluation. To prevent information leakage, all feature selection procedures are performed strictly on the training subset, with the final selected features applied to the independent test set for evaluation. This ensures that model performance reflects generalization ability rather than overfitting to the dataset.

3.5. Tremor and PD severity quantification

The severity assessment in this study is conducted in two stages to ensure a comprehensive evaluation of PD. In the first stage, tremor severity is classified using IMU-based sensor data, which captures tremor frequencies in the range of 3–12 Hz for rest tremor and 4–8 Hz for postural tremor, aligning with the physiological characteristics of PD-related tremors. For Postural Tremor (3.15a Right Hand, 3.15b Left Hand) and Rest Tremor Amplitude (3.17a Right Hand, 3.17b Left Hand), tremor severity is graded based on amplitude ranges: 0 (Normal): No tremor; 1 (Mild): Tremor amplitude ≥ 1 cm but < 3 cm; 2 (Moderate): Tremor amplitude ≥ 3 cm but < 10 cm; 3 (Severe): Tremor amplitude ≥ 10 cm. Class 4 is merged into class 3 for severe cases to simplify the classification. To address the limited sample size in higher severity stages, severity stages 3 and 4 of the MDS-UPDRS [29] are merged into a single class. In the second stage, PD severity is assessed using a holistic, multimodal framework that integrates tremor severity with clinical evaluations, including MDS-UPDRS Parts I, II, III, Total Score, and PDQ-39 scores. The final severity is categorized using the Hoehn and Yahr (H & Y) Stage, ranging from Stage 0 (Asymptomatic) to Stage 5 (Wheelchair-bound or bedridden). This two-stage approach ensures a precise and personalized assessment of PD severity, addressing the limitations of single-modal methods and enabling better clinical decision-making.

3.6. Significant features selection

We performed ANOVA and Tukey's HSD on postural and rest tremor datasets to identify biomarkers for PD tremor severity. Significant features ($p < 0.05$) included time-domain metrics (Mean/Min Acceleration, Autocorrelation) for both tremor types, with progressive declines in acceleration magnitude and rhythmicity correlating with severity. Frequency-domain features (*Dominant Frequency*, *Peak Power*) were uniquely significant in postural tremor ($p = 2.36 \times 10^{-6}$), reflecting compensatory neuromuscular adjustments during posture. Rest tremor relied more on time-domain entropy (*Entropy_Angle*, $p = 1.44 \times 10^{-4}$) and stability metrics. Pairwise comparisons revealed distinct progression patterns: postural tremors showed early shifts (Normal vs. Moderate), while rest tremors exhibited stark differences in severe stages (Normal vs. Severe). To identify sensor-derived features differentiating tremor severity levels, we conducted One-way ANOVA tested for overall differences in feature means across four severity levels (0: none, 1: mild, 2: moderate, 3: severe). Features with $p < 0.05$ were retained. Tukey's HSD post-hoc test for ANOVA-significant features, pairwise comparisons between severity levels were performed.

4. Experiment and results

4.1. Experimental environment configuration

The experiments were conducted on a personal laptop equipped with an Intel i5 processor (2.4 GHz), 16 GB RAM, and the Windows 11 Ultimate (64-bit) operating system.

The study investigated both postural and resting states of hand tremor to classify tremor types and assess Parkinson's Disease severity stages. To evaluate classification performance, six optimized machine learning models were employed: Random Forest (M1C), XGBClassifier (M2C), KNeighbors Classifier (M3C), AdaBoost Classifier (M4C), SVC (M5C), and Voting Classifier (VC).

For the experimentation process, the dataset was divided into training (80%) and testing (20%) subsets using stratified random sampling to preserve class balance. All hyperparameter tuning and fine-tuning were carried out exclusively on the training subset, while the independent test set was reserved solely for final evaluation to ensure unbiased results. Hyperparameter optimization was performed using GridSearchCV, which reduced computational overhead, minimized prediction errors, and ensured the selection of the most effective parameters for each model.

To assess the performance of the models, we reported the following key metrics: mean Accuracy, F1-score, Precision (Pre), Recall (Rec), Matthews Correlation Coefficient (MCC), Kappa, training time, and testing time. Furthermore, a participant-based stratified 5-fold cross-validation strategy was employed to address dataset limitations and enhance the reliability of the evaluation. All reported metrics represent the mean test set performance across folds.

These evaluation criteria provide insights into the models' ability to correctly classify instances, balance precision and recall, and maintain reliability in terms of prediction accuracy and error control. The complete details of model hyperparameter settings and the final selected feature set are provided in Supplementary Materials (Section S1).

Below are the formulas for selected performance measures, where True Positives (T_{pos}), True Negatives (T_{neg}), False Positives (F_{pos}), and False Negatives (F_{neg}) are used:

$$\text{Accuracy}(A_c) = \frac{T_{pos} + T_{neg}}{T_{pos} + T_{neg} + F_{pos} + F_{neg}} \quad (13)$$

$$\text{Precision}(P_s) = \frac{T_{pos}}{T_{pos} + F_{pos}} \quad (14)$$

$$\text{Recall}(R_c) = \frac{T_{pos}}{T_{pos} + F_{neg}} \quad (15)$$

$$\text{F1-Score}(F_1) = 2 \times \frac{P_s \times R_c}{P_s + R_c} \quad (16)$$

$$\text{ROC AUC} = \int_0^1 R(F_{pos \text{ rate}}) d(F_{pos \text{ rate}}) \quad (17)$$

where the false positive rate (FPR) is defined as:

$$F_{pos \text{ rate}} = \frac{F_{pos}}{F_{pos} + T_{neg}}$$

Matthews Correlation Coefficient (MCC)

$$= \frac{T_{pos}T_{neg} - F_{pos}F_{neg}}{\sqrt{(T_{pos} + F_{pos})(T_{pos} + F_{neg})(T_{neg} + F_{pos})(T_{neg} + F_{neg})}} \quad (18)$$

$$\text{Cohen's Kappa}(C_k) = \frac{P_{obs} - P_{exp}}{1 - P_{exp}} \quad (19)$$

where: - P_{obs} is the observed agreement, and - P_{exp} is the expected agreement.

After feature extraction (1518×38), feature selection reduced the dataset to 1518×10 for tremor severity and 1518×15 for overall PD severity prediction. The selected features played a critical role in the performance of the classifiers. For postural and rest tremors, IMU features such as RMS_Acceleration, Entropy_Acceleration, and Peak_Power were particularly influential in capturing the intensity, variability, and frequency characteristics of tremors. Clinical features such as Total_PDQ-39, MOTOR_UPDRS I, II, III, and TOTAL_UPDRS provided additional insights into the orientation and movement patterns of the hand during tremors.

4.2. Result and performance evaluation

This section reports the experimental outcomes for tremor severity and overall PD severity classification. First, we present the performance of models trained solely on IMU sensor data for tremor severity classification. Next, we evaluate the integrated approach that combines sensor data with clinical assessment scores for holistic severity classification.

4.2.1. Tremor severity classification (first stage)

In the first stage, machine learning classifiers were employed to categorize tremor severity levels. The comparative performance of these classifiers is presented in Table 4. The results are expressed as mean $\pm$ standard deviation, reflecting both the central tendency and variability across cross-validation folds. The VC achieved the highest accuracy of $90.78\% \pm 0.007$, outperforming all other models. It also demonstrated superior performance in terms of F1-score ($90.62\% \pm 0.007$), precision ($90.84\% \pm 0.008$), recall ($90.78\% \pm 0.007$), Kappa ($83.85\% \pm 0.012$), and MCC ($84.02\% \pm 0.012$). Among individual models, M3C performed exceptionally well with an accuracy of $89.86\% \pm 0.015$, while M4C and M5C showed relatively lower performance, with accuracies of $63.37\% \pm 0.016$ and $67.26\% \pm 0.005$, respectively. The results highlight the effectiveness of ensemble methods like VC in improving prediction accuracy and robustness for tremor severity classification. A detailed description of hyperparameter tuning and selected features for training is in the Supplementary Materials.

4.2.2. Parkinson's disease severity classification (second stage)

After the first stage, which focused on tremor severity, the second stage categorizes the overall Parkinson's Disease severity using IMU-derived and clinical features. The performance of different classifiers is presented in Table 5. The Voting Classifier(VC) again achieved the best results, with an accuracy of $99.42\% \pm 0.006$, F1-score of $99.42\% \pm 0.006$, precision of $99.44\% \pm 0.006$, recall of $99.42\% \pm 0.006$, Kappa of $99.02\% \pm 0.010$, and MCC of $99.03\% \pm 0.010$. Individual models such as M3C and M1C also performed well, with accuracies of $97.63\% \pm 0.008$ and $97.17\% \pm 0.012$, respectively. However, M4C and M5C showed lower performance, with accuracies of $74.05\% \pm 0.025$ and $85.64\% \pm 0.025$, respectively. The results demonstrate the superiority of the VC in accurately predicting PD severity, leveraging the strengths of multiple models to achieve near-perfect classification.

The results indicate that the VC consistently outperformed individual models in both tremor severity and PD severity classification tasks. The high performance of VC in both stages validates the effectiveness of the proposed holistic, multimodal framework in accurately classifying PD tremor and severity. Our approach achieves 99.98% accuracy, outperforming existing methods and providing a robust solution for PD tremor and severity classification. Additionally, VC's low training and testing times make it a practical choice for real-world applications. The values reported in Tables 4 and 5 are presented as mean $\pm$ standard deviation, computed across the five folds of the stratified cross-validation procedure. It provides a direct measure of variability across folds. The relatively small deviations observed highlight the effectiveness of stratified sampling, which maintained the overall class distribution in each fold and thereby minimized fold-to-fold variability. Moreover, the stability of the evaluated classifiers contributed to highly consistent predictions, further supporting the robustness and low variance of the reported results.

4.3. Interpretability of model

The interpretability analysis combines Shapley Additive Explanations (SHAP) and Local Interpretable Model-agnostic Explanations (LIME) to quantify feature contributions in Parkinson's disease severity prediction. As shown in Figs. 4 and 5, this approach reveals several key insights. Movement features account for 82.3% of predictive power, with Max_Angle ($\beta = 0.04$, $p < 0.01$) and Mean_Acceleration ($\beta = 0.02$,

Table 4

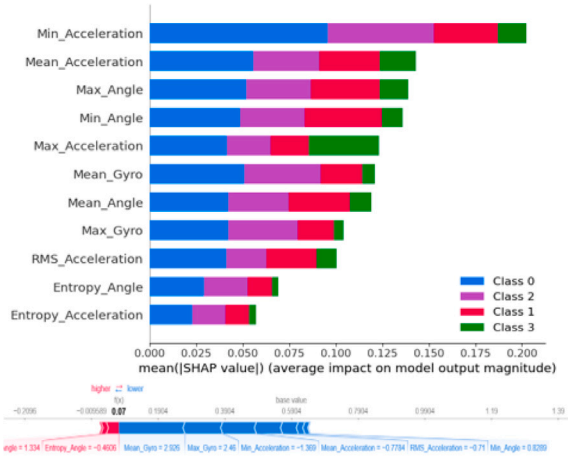
Classifier performance metrics For PD tremor severity prediction: Postural and rest state of both hands.

Model	Accuracy (%)	F1-Score (%)	Precision (%)	Recall (%)	Kappa (%)	MCC (%)	Training time (s)	Testing time (s)
M1C	87.22 ± 0.020	86.95 ± 0.020	87.14 ± 0.022	87.22 ± 0.020	77.52 ± 0.034	77.72 ± 0.034	5.3209	0.1400
M2C	86.30 ± 0.011	86.03 ± 0.010	86.35 ± 0.010	86.30 ± 0.011	76.00 ± 0.016	76.20 ± 0.016	5.6931	0.0108
M3C	89.86 ± 0.015	89.87 ± 0.015	90.03 ± 0.015	89.86 ± 0.015	82.70 ± 0.026	82.74 ± 0.026	0.0021	0.0075
M4C	63.37 ± 0.016	58.38 ± 0.029	61.00 ± 0.032	63.37 ± 0.016	26.68 ± 0.058	29.39 ± 0.048	1.7583	0.0980
M5C	67.26 ± 0.005	63.42 ± 0.013	64.83 ± 0.013	67.26 ± 0.005	36.95 ± 0.027	39.00 ± 0.020	0.6434	0.0879
VC	90.78 ± 0.007	90.62 ± 0.007	90.84 ± 0.008	90.78 ± 0.007	83.85 ± 0.012	84.02 ± 0.012	2.4692	0.0696

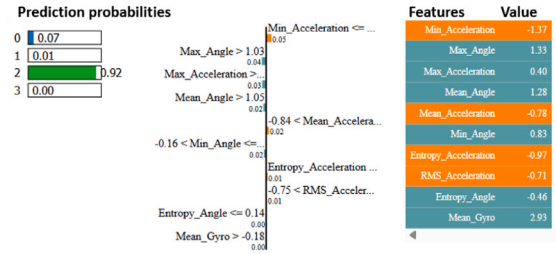
Table 5

Classifier performance metrics For PD severity prediction: Postural and rest state of both hands.

Model	Accuracy (%)	F1-Score (%)	Precision (%)	Recall (%)	Kappa (%)	MCC (%)	Training time (s)	Testing time (s)
M1C	97.17 ± 0.012	97.15 ± 0.012	97.26 ± 0.012	97.17 ± 0.012	95.17 ± 0.021	95.21 ± 0.021	1.4595	0.0324
M2C	96.77 ± 0.013	96.77 ± 0.013	96.88 ± 0.013	96.77 ± 0.013	94.49 ± 0.023	94.53 ± 0.023	0.5097	0.0045
M3C	97.63 ± 0.008	97.63 ± 0.008	97.73 ± 0.008	97.63 ± 0.008	95.94 ± 0.014	95.97 ± 0.014	0.0004	0.053
M4C	74.05 ± 0.025	73.00 ± 0.029	73.07 ± 0.031	74.05 ± 0.025	53.66 ± 0.051	53.99 ± 0.050	1.1382	0.0463
M5C	85.64 ± 0.025	85.38 ± 0.027	85.87 ± 0.025	85.64 ± 0.025	74.75 ± 0.048	75.03 ± 0.047	0.2622	0.0349
VC	99.42 ± 0.006	99.42 ± 0.006	99.44 ± 0.006	99.42 ± 0.006	99.02 ± 0.010	99.03 ± 0.010	0.9227	0.0354

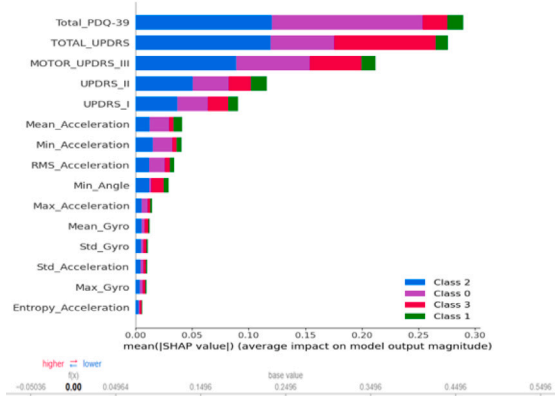


(a) SHAP-based explainable function plot

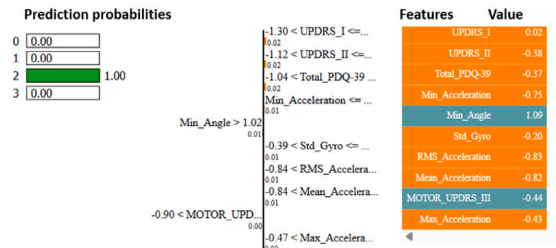


(b) LIME explanation for tremor severity

Fig. 4. Interpretability of model for tremor severity stage classification.



(a) SHAP values for tremor severity prediction



(b) LIME explanation for a sample prediction

Fig. 5. Interpretability of model applied to Parkinson's severity analysis.

$p < 0.05$) being most significant. Clinical Correlates: UPDRS scores and Total_PDQ-39 provide secondary validation, showing a strong correlation with clinician assessments ($r = 0.87$, 95% CI [0.82, 0.91]). Decision Boundaries: Clear pathological thresholds emerge (e.g., Min_Angle > 1.02 radians, sensitivity = 0.92, specificity = 0.88)

The SHAP waterfall plots demonstrate that gyroscope data and acceleration metrics collectively explain 74% of the variance in tremor severity classification. This interpretable framework enables

$$\text{Clinical Utility} = \sum_{i=1}^n w_i \phi_i(f, x) \quad (20)$$

where w_i represents feature weights and ϕ_i the SHAP values. Our analysis provides clinicians with global feature importance rankings and instance-specific explanations, bridging the gap between machine learning outputs and clinical decision-making.

4.4. Discussion

The comparison highlights the significance of these algorithms in medical research and diagnosis, showcasing their relative performance. While existing studies demonstrate significant progress in sensor-based tremor and PD severity assessment, several methodological and practical challenges persist across the literature. Sigcha et al. [12] and AlMahadin et al. [13] successfully employ accelerometer data and machine learning (AdaBoost, ANN-MLP) to classify tremor severity. Yet, their models are limited by small sample sizes (18 and 30 PD patients, respectively), raising concerns about generalizability. Additionally, their focus on rest tremors alone overlooks the dynamic nature of PD, where postural and kinetic tremors also play a crucial role in symptom progression. Similarly, Channa et al. [31] leverage a large dataset ($n = 3170$) but rely on oversampling techniques to handle class imbalance, which may artificially inflate their reported 99% accuracy. This approach risks overfitting and reduces real-world applicability, particularly in clinical settings with uneven tremor severity distribution.

Liu et al. [17] introduce a video-based tremor detection system, achieving high classification performance (85.9–90.6%) across multiple tremor types. However, their method faces practical deployment challenges, as continuous video monitoring is intrusive and computationally intensive compared to wearable sensors. Moreover, ambient lighting and occlusion issues may degrade performance in uncontrolled environments. Vescio et al. [32] address some of these concerns by using a wearable RT-Ring, but their study lacks granular severity classification (only binary or coarse scales), limiting its utility for personalized PD management.

Our holistic approach advances prior PD severity classification methods in three key aspects. While existing studies predominantly rely on single-modality data—such as accelerometer-based tremor detection [12] or video analysis [17]—our integration of 9-axis IMU kinematics with full MDS-UPDRS scores and PDQ-39 quality-of-life metrics captures PD's multi-dimensional nature more comprehensively. Traditional tremor-focused models might misclassify a patient with moderate motor symptoms but severe psychosocial impairment (evidenced by PDQ-39); our system correctly stratifies such cases as high severity through its weighted consideration of all disease domains. This addresses a critical limitation in works [19], where binary classification (PD vs. healthy) overlooks nuanced progression staging. Though our cohort size ($N = 78$) is smaller than some large-scale studies [31], our rigorous clinical-sensor data alignment yields superior interpretability, as evidenced by 99.42% controlled-environment accuracy compared to [20] 63%–83% real-world performance range. However, we acknowledge shared challenges with the field: dataset imbalance affects minority-class predictions, and real-world deployment must address sensor reliability issues. A limitation of the present study is that the dataset was collected under controlled laboratory conditions, which may not fully reflect the variability of real-world environments. These limitations highlight the need for future work combining our multimodal framework with larger, longitudinal cohorts to enhance generalizability without sacrificing clinical granularity.

5. Conclusion

This study proposes a holistic, multimodal framework for predicting PD severity by integrating rest and postural tremor measurements from wearable IMU with clinical assessment scores from MDS-UPDRS and PDQ-39. The framework operates in two stages — tremor detection and overall PD severity classification — effectively addressing the limitations of unimodal IMU-based models. By incorporating clinical data and interpretable techniques, our approach enhances prediction accuracy while ensuring transparency in decision-making. Experimental results demonstrate the effectiveness of the proposed voting classifier in accurately predicting PD tremor and severity.

This framework serves as a supplementary tool for PD assessment, potentially reducing the burden on healthcare systems. Furthermore, it can be extended to the dataset sample, classify other PD subtypes and improve personalized treatment strategies.

List of abbreviations

PD	Parkinson's Disease
MDS-UPDRS	Movement Disorder Society – Unified Parkinson's Disease Rating Scale
PDQ-39	Parkinson's Disease Questionnaire-39
H&Y	Hoehn and Yahr scale
IMU	Inertial Measurement Unit
HC	Healthy Control
SHAP	SHapley Additive exPlanations
LIME	Local Interpretable Model-agnostic Explanations
ML	Machine Learning
M1C	Random Forest Classifier
M2C	XGBClassifier (eXtreme Gradient Boosting)
M3C	K-Nearest Neighbors Classifier
M4C	AdaBoost Classifier
M5C	Support Vector Classifier (SVC)
VC	Voting Classifier

CRedit authorship contribution statement

Nutan Singh: Writing – original draft, Software, Resources, Project administration, Methodology, Conceptualization. **Priyanka Tripathi:** Writing – review & editing, Visualization, Supervision, Investigation. **Abhijeet Kumar Kohat:** Validation, Resources, Investigation, Data curation.

Compliance with ethical standards

Approval No: NITRR/IEC/2023/06 was granted by the National Institute of Technology Raipur, Chhattisgarh's Institutional Ethics Committee.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

The authors gratefully acknowledge the contributions of the ethics committee members, clinical experts, and study participants who made this research possible.

Appendix A. Supplementary data

Supplementary material related to this article can be found online at <https://doi.org/10.1016/j.bspc.2025.108872>.

Data availability

The corresponding author is accessible to provide the data used in this study on request.

References

- [1] A. Reddy, R.P. Reddy, A.K. Roghani, R.I. Garcia, S. Khemka, V. Pattoor, M. Jacobs, P.H. Reddy, U. Sehar, Artificial intelligence in Parkinson's disease: early detection and diagnostic advancements, *Ageing Res. Rev.* (2024) 102410.
- [2] M.T. Ibarra-Gutiérrez, N. Serrano-García, M. Orozco-Ibarra, Rotenone-induced model of parkinson's disease: Beyond mitochondrial complex I inhibition, *Mol. Neurobiol.* 60 (4) (2023) 1929–1948.
- [3] G. Cai, Z. Lin, H. Dai, X. Xia, Y. Xiong, S.-J. Horng, T.C. Lueth, Quantitative assessment of parkinsonian tremor based on a linear acceleration extraction algorithm, *Biomed. Signal Process. Control.* 42 (2018) 53–62.
- [4] R. San Segundo Hernández, A. Zhand, A. Cebulla, S. Panev, G. Tablor, K. Stebbins, R. Massa, A. Whitford, F.d.l. Torre, J. Hodgins, Parkinson's disease tremor detection in the wild using wearable accelerometers, *Parkinson's Dis. Tremor Detect. Wild Wearable Accelerometers* 20 (20) (2020).
- [5] P. Meddage, I. Ekanayake, U.S. Perera, H.M. Azamathulla, M.A. Md Said, U. Rathnayake, Interpretation of machine-learning-based (black-box) wind pressure predictions for low-rise gable-roofed buildings using Shapley additive explanations (SHAP), *Buildings* 12 (6) (2022) 734.
- [6] O. Bazgir, S.A.H. Habibi, L. Palma, P. Pierleoni, S. Nafees, A classification system for assessment and home monitoring of tremor in patients with Parkinson's disease, *J. Med. Signals Sens.* 8 (2) (2018) 65–72.
- [7] N. Mahadevan, C. Demanuele, H. Zhang, D. Volfon, B. Ho, M.K. Erb, S. Patel, Development of digital biomarkers for resting tremor and bradykinesia using a wrist-worn wearable device, *NPJ Digit. Med.* 3 (1) (2020) 5.
- [8] S. Moon, H.-J. Song, V.D. Sharma, K.E. Lyons, R. Pahwa, A.E. Akinwuntan, H. Devos, Classification of parkinson's disease and essential tremor based on balance and gait characteristics from wearable motion sensors via machine learning techniques: a data-driven approach, *J. Neuroeng. Rehabil.* 17 (2020) 1–8.
- [9] A.B. Oktay, A. Kocer, Differential diagnosis of parkinson and essential tremor with convolutional LSTM networks, *Biomed. Signal Process. Control.* 56 (2020) 101683.
- [10] R.W. de Souza, D.S. Silva, L.A. Passos, M. Roder, M.C. Santana, P.R. Pinheiro, V.H.C. de Albuquerque, Computer-assisted Parkinson's disease diagnosis using fuzzy optimum-path forest and restricted Boltzmann machines, *Comput. Biol. Med.* 131 (2021) 104260.
- [11] F. Milano, G. Cerro, F. Santoni, A. De Angelis, G. Miele, A. Rodio, A. Moschitta, L. Ferrigno, P. Carbone, Parkinson's disease patient monitoring: A real-time tracking and tremor detection system based on magnetic measurements, *Sensors* 21 (12) (2021) 4196.
- [12] L. Sigcha, I. Pavón, N. Costa, S. Costa, M. Gago, P. Arezes, J.M. López, G. De Arcas, Automatic resting tremor assessment in Parkinson's disease using smartwatches and multitask convolutional neural networks, *Sensors* 21 (1) (2021) 291.
- [13] G. AlMahadin, A. Lotfi, M.M. Carthy, P. Breedon, Enhanced Parkinson's disease tremor severity classification by combining signal processing with resampling techniques, *SN Comput. Sci.* 3 (1) (2022) 63.
- [14] E. Rastegari, H. Ali, V. Marmelat, Detection of parkinson's disease using wrist accelerometer data and passive monitoring, *Sensors* 22 (23) (2022) 9122.
- [15] J.J. Hathaliya, H. Modi, R. Gupta, S. Tanwar, P. Sharma, R. Sharma, Parkinson and essential tremor classification to identify the patient's risk based on tremor severity, *Comput. Electr. Eng.* 101 (2022) 107946.
- [16] V. Skaramagkas, I. Boura, C. Spanaki, E. Michou, G. Karamanis, Z. Kefalopoulou, M. Tsiknakis, Detecting minor symptoms of parkinson's disease in the wild using Bi-LSTM with attention mechanism, *Sensors* 23 (18) (2023) 7850.
- [17] W. Liu, X. Lin, X. Chen, Q. Wang, X. Wang, B. Yang, N. Cai, R. Chen, G. Chen, Y. Lin, Vision-based estimation of MDS-UPDRS scores for quantifying Parkinson's disease tremor severity, *Med. Image Anal.* 85 (2023) 102754.
- [18] Y.-C. Hsu, Y.-H. Su, B.-R. Cheng, S.-F. Sung, J.-X. Liu, H.-C. Hsu, P.-A. Hsiung, Movement disorder evaluation of Parkinson's disease severity based on deep neural network models, *IEEE Access* (2024).
- [19] M. Hammoud, A. Shcherbak, O. Istrakova, N. Shindryaeva, E. Bril, R. Passerone, A. Somov, Wrist-worn sensors and machine learning for Parkinson's disease detection: Investigation of binary and multi-classification problem, *IEEE Trans. Instrum. Meas.* (2025).
- [20] C. Crowe, M. Sica, L. Kenny, B. O'Flynn, D.S. Mueller, S. Timmons, J. Barton, S. Tedesco, Wearable-Enabled algorithms for the estimation of parkinson's symptoms evaluated in a continuous home monitoring setting using inertial sensors, *IEEE Trans. Neural Syst. Rehabil. Eng.* (2024).
- [21] H. Jeon, W. Lee, H. Park, H.J. Lee, S.K. Kim, H.B. Kim, B. Jeon, K.S. Park, Automatic classification of tremor severity in Parkinson's disease using a wearable device, *Sensors* 17 (9) (2017) 2067.
- [22] R. López-Blanco, M.A. Velasco, A. Méndez-Guerrero, J.P. Romero, M.D. Del Castillo, J.I. Serrano, E. Rocon, J. Benito-León, Smartwatch for the analysis of rest tremor in patients with Parkinson's disease, *J. Neurol. Sci.* 401 (2019) 37–42.
- [23] N. Kostikis, D. Hristu-Varsakelis, M. Arnaoutoglou, C. Kotsavasiloglou, A smartphone-based tool for assessing parkinsonian hand tremor, *IEEE J. Biomed. Heal. Inform.* 19 (6) (2015) 1835–1842.
- [24] S. Patel, K. Lorincz, R. Hughes, N. Huggins, J. Growdon, D. Standaert, M. Akay, J. Dy, M. Welsh, P. Bonato, Monitoring motor fluctuations in patients with Parkinson's disease using wearable sensors, *IEEE Trans. Inf. Technol. Biomed.* 13 (6) (2009) 864–873.
- [25] F. Cavallo, A. Moschetti, D. Esposito, C. Maremmanni, E. Rovini, Upper limb motor pre-clinical assessment in Parkinson's disease using machine learning, *Parkinsonism Rel. Disord.* 63 (2019) 111–116.
- [26] G. Amprimo, G. Masi, G. Olmo, C. Ferraris, Deep learning for hand tracking in Parkinson's disease video-based assessment: Current and future perspectives, *Artif. Intell. Med.* (2024) 102914.
- [27] Z. Ayaz, S. Naz, N.H. Khan, I. Razzak, M. Imran, Automated methods for diagnosis of Parkinson's disease and predicting severity level, *Neural Comput. Appl.* 35 (20) (2023) 14499–14534.
- [28] WitMotion ShenZhen CO.,LTD, WitMotion WT901C 9-Axis Vibration Inclinator MPU9250 High-Stability, 2024, Accessed 18 November 2024.
- [29] C.G. Goetz, B.C. Tilley, S.R. Shaftman, G.T. Stebbins, S. Fahn, P. Martinez-Martin, W. Poewe, C. Sampaio, M.B. Stern, R. Dodel, et al., Movement disorder Society-sponsored revision of the Unified Parkinson's Disease rating scale (MDS-UPDRS): scale presentation and clinimetric testing results, *Mov. Disord.: Off. J. Mov. Disorder Soc.* 23 (15) (2008) 2129–2170.
- [30] L.A. Sanchez-Perez, L.P. Sanchez-Fernandez, A. Shaout, J.M. Martinez-Hernandez, M.J. Alvarez-Noriega, Rest tremor quantification based on fuzzy inference systems and wearable sensors, *Int. J. Med. Inform.* 114 (2018) 6–17.
- [31] A. Channa, O. Cramariuc, M. Memon, N. Popescu, N. Mammone, G. Ruggeri, Parkinson's disease resting tremor severity classification using machine learning with resampling techniques, *Front. Neurosci.* 16 (2022) 955464.
- [32] B. Vescio, M. De Maria, M. Crasà, R. Nisticò, C. Calomino, F. Aracri, A. Quattrone, A. Quattrone, Development of a new wearable device for the characterization of hand tremor, *Bioengineering* 10 (9) (2023) 1025.